SUPPORTING ADULT GUIDANCE

Cross-sections

2D and 3D Shapes. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner meets the idea of a **cross-section** — the 2D shape revealed by slicing a solid — and the definition of a **prism** as a solid with a constant cross-section along its length. They slice cuboids, prisms, pyramids, cylinders and cones, and notice how the **direction** of a cut changes the shape. The emphasis is on **spatial reasoning**: seeing the 2D shape hidden inside the 3D object.

Driving Question: *What shape appears when you slice straight through a solid?*

The mathematics, in plain terms

A **cross-section** is the flat 2D shape you get when you slice straight through a solid. The most important idea is the **prism**: a solid that has the **same cross-section all the way along its length**. A cuboid, a triangular prism and a cylinder are all prisms — every level slice is identical (a rectangle, a triangle, a circle). This is why, in the next lesson, the volume of any prism is simply its **cross-section area × length**. Solids that are **not** prisms — a pyramid or a cone — have a cross-section that **shrinks** as you move toward the point. The shape of a slice also depends on the **direction** of the cut: a cylinder sliced level gives a circle, sliced straight down gives a rectangle, and sliced at a slant gives an ellipse. A cone is the richest of all: level slices are circles, and slanted cuts reveal an ellipse or a parabola. None of this needs calculation — it is about picturing the 2D shape inside the 3D solid, which the interactive makes visible.

The 2D shape inside the 3D solid

The heart of this lesson is a shift of imagination: learning to see the flat shape hidden inside a solid, and to picture how a cutting plane meets it. The most useful thing to protect is that the learner reaches the idea of the constant cross-section themselves — by carrying a shape along to build a prism, and by sliding a slice up a prism and watching it stay the same. Once that is secure, the contrast with a pyramid or cone (whose slice shrinks) does the defining work: a prism is exactly the solid whose slices do not change. The tilt slider and the cone are enrichment — a first, memorable glimpse of ellipses and parabolas — not something to be assessed. This lesson builds straight into Volume (L9), where a prism's volume is cross-section area × length.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Extruder: carry a 2D cross-section along a length to build a prism, and see that every slice is the same.	~12 min

Movement	The Slicing Plane Studio: slice any solid at any height and tilt, and see the cross-section face-on.	~12 min
Reflection	Name the Slice: predict the 2D shape a cut will reveal, and reason about why.	~9 min
Creativity	Design your own prism from a chosen cross-section, and predict a slice halfway along.	~5 min
Rest	A pause between dimensions — noticing which slice surprised you most.	~4 min
Nutrition	Contextual (Mode A): sliced bread, CT/MRI body scans, architectural sections and cut fruit.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking every slice of a solid is the same, whatever the solid

Only prisms have a constant cross-section. Repair: “Slide the cut up a pyramid — does the square stay the same size, or shrink?”

Thinking the cross-section depends only on the solid, not on the cut direction

Repair: “Slice this cylinder level, then straight down, then on a slant — do you get the same shape each time?”

Believing a cylinder always gives a circle

A vertical cut gives a rectangle; a slanted cut gives an ellipse. Repair: “Tilt the slice — what happens to the circle?”

Confusing a cross-section (a 2D slice) with a net or a face

A face is the outside surface; a cross-section is revealed by cutting through the inside. Repair: “Are we looking at the outside, or at what a cut reveals inside?”

Assuming any solid with a uniform look is a prism

A prism needs the same cross-section end to end. Repair: “Could you slice it anywhere along its length and always get the same shape and size?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the names and properties of basic 2D shapes (L1) and the idea of a solid's faces. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the picturing — and the reasoning — with the learner:

- “Picture the cut. Which edges does the slicing plane cross?”
- “What 2D shape do those crossing points make?”
- “If you moved the cut higher, would the shape stay the same, or change?”
- “Is this solid a prism? Could you slice it anywhere along its length and always get the same shape?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Do they need to memorise which slice gives which shape?”

No — the point is to reason it out by picturing the cut, which the Slicing Plane Studio makes visible. Recognising a prism by its constant cross-section matters far more than memorising a list.

“Why does the prism idea matter so much?”

It is the bridge to the next lesson: the volume of any prism is its cross-section area × length. Securing ‘same cross-section all along’ now makes volume feel obvious later.

“The ellipse and parabola from the cone look advanced.”

They are enrichment — a first, memorable encounter, not something to be tested. Let the learner enjoy discovering that one solid can hide so many curves; the core learning is the prism and the idea of a slice.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like

Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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